

EXP. NO: 9

PERFORMANCE EVALUATION OF AM GENERATION AND DETECTION UNDER NOISY CHANNEL CONDITIONS USING

MATLAB

Objectives:

- i) To generate an Amplitude Modulated (AM) signal, introduce Additive White Gaussian Noise (AWGN), and analyze the effect of noise on the transmitted signal using MATLAB.
- ii) To recover the original message signal from the noisy AM signal using envelope detection and compare the transmitted and recovered waveforms to evaluate the performance of the AM communication system.

CODE

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;
SNR = 10;           % Signal-to-noise ratio in dB

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);

% Add AWGN noise (manual implementation)
signalPower = mean(am.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower)*randn(size(am));
```

```

noisy = am + noise;

% Envelope Detection (rectification)
env = abs(noisy);

% Smooth the envelope
window = round(fs/(2*fc));
env = movmean(env, window);

% Remove DC offset
recovered = env - mean(env);

% Normalize recovered signal
recovered = recovered / max(abs(recovered));

% Display Results
figure('Name','AM with AWGN Noise','NumberTitle','off');

subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

subplot(5,1,3);
plot(t,am);
title('AM Signal');
grid on;

subplot(5,1,4);
plot(t,noisy);
title('AM Signal with AWGN Noise (10 dB SNR)');
grid on;

subplot(5,1,5);
plot(t,recovered);
title('Recovered Message Signal');
xlabel('Time (s)');
grid on;

```

OUTPUT

